Week 3 Narrative Report: Synesthetic Learning Project

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This week, our work shifted toward building the first functional components of the Synesthetic Learning model. With the datasets already cleaned and aligned from previous weeks, the focus now was on defining how the system will process audio and begin learning emotional patterns that will later link to abstract art.

My contribution centered on setting up the initial modeling pipeline. I worked on the audio preprocessing functions that convert raw files into Mel-spectrograms, ensuring every sample was standardized and prepared for training. I also helped evaluate several model options and contributed to selecting a simple CNN-based classifier as our starting baseline.

To test the setup, I ran a small preliminary training session using a subset of the audio data. This allowed us to confirm that the spectrogram pipeline, batching system, and training loop were all running correctly. The model showed reasonable early learning behavior, which gave us confidence to proceed with full-scale experiments.

I also drafted the outline for our basic retrieval system, where predicted audio emotions will later be paired with artworks that share the same label. This baseline design prepares the project for the cross-modal embedding phase planned for the coming weeks.

Overall, I focused on helping establish a stable and working audio classification pipeline, testing its feasibility, and supporting the early decisions that guide the next steps in model development.